

E-Commerce Transactions — Insight Report

AnalystLab Africa Data Analytics Internship | Week 1–2

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1. Dataset Overview

Metric	Value
Source	UCI Machine Learning Repository — Online Retail Dataset
Total Records (Raw)	541,909 rows
Total Records (Cleaned)	524,878 rows
Columns	8 (InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country)
Date Range	December 01, 2010 – December 09, 2011
Unique Customers	4,338
Unique Products	4,015
Countries Covered	38

2. Data Cleaning Summary

Issue Identified	Action Taken
5,268 duplicate rows	Removed using drop_duplicates()
1,454 missing Descriptions	Rows dropped — no business value without product name
135,080 missing CustomerIDs	Filled with 0 to represent guest/unknown customers
Negative Quantity values (returns)	Separated and removed from main dataset
Negative & zero UnitPrice anomalies	Removed — invalid for revenue analysis
InvoiceDate stored as text (object)	Converted to datetime format
CustomerID stored as float	Converted to integer
Inconsistent text formatting	Standardized to Title Case for Description & Country

3. Key Insights

Insight 1 — UK Dominates Revenue

The United Kingdom is overwhelmingly the top revenue-generating country, accounting for the vast majority of the \$10,642,110.80 total revenue. This indicates heavy market concentration and highlights international expansion as a key growth opportunity.

Insight 2 — Sales are Strongly Seasonal

November 2011 was the best performing sales month, consistent with pre-Christmas shopping and Black Friday demand. The business should prioritize inventory buildup and targeted marketing campaigns ahead of Q4 each year.

Insight 3 — Small Customer Base Drives Large Revenue

Only 4,338 unique customers generated over \$10.6 million in revenue — an average spend of approximately \$2,452 per customer. This signals a high-value, loyal customer base where retention strategies would yield significant returns.

Insight 4 — Paper Craft Little Birdie is the Top Product

This product recorded the highest quantity sold across the entire dataset. Craft and gift items dominate demand, suggesting the retailer should ensure consistent stock availability for this product category.

Insight 5 — Business Faces Concentration Risk

With revenue heavily concentrated in the UK market and among a relatively small customer pool, the business is exposed to concentration risk. Diversifying into new markets and acquiring new customers should be a strategic priority.